

Haoan Feng

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ABOUT ME

I am a researcher focused on the intersection of computer vision, geometry processing, and geospatial data analysis. My research centers on neural representations of geospatial data, combining implicit neural modeling, topology-aware analysis, and generative frameworks to enable interpretable, continuous, and scalable representations of the physical world. I am motivated by interdisciplinary collaboration that bridges machine learning, graphics, and scientific computing.

WORK EXPERIENCE

PhD Research Intern at Dolby ATG Laboratories 05/2026 - 08/2026
Sunnyvale, CA Mentors: Peimeng Guan & Dae Yeol Lee, Manager: Guan-Ming Su

- ◇ Proposed and led a greenfield research project on authoring text and icon annotations as native TVGS Gaussian assets inside volumetric video, achieving correct mutual occlusion and physically based relighting in a single rendering pass, with a Viser-based interactive scene editor for keyframe and text-track authoring.
- ◇ Designed a feed-forward amortized authoring method where a shared encoder-decoder predicts renderable assets directly from canonical anchor point clouds, keeping normal map accuracy under 2 degrees while shrinking per-asset storage by roughly 28×.
- ◇ Built the multi-GPU training stack and scaled it to a library of 27K+ physically based assets that generalizes to fonts never seen in training.

PhD Research Intern at Dolby ATG Laboratories 05/2025 - 08/2025
Sunnyvale, CA Manager: Guan-Ming Su

- ◇ Developed a geometry-guided framework to augment Video Large Language Models (VideoLLMs) with cinematographic camera motion understanding, resulting in a first-author publication accepted at the CVPR 2026 Workshop (PVUW).
- ◇ Created a synthetic video benchmark using Unreal Engine 5 and designed probing experiments to evaluate spatial retention in vision encoders via Q-former.
- ◇ Proposed a plug-and-play strategy that injects camera motion, framing, and spatial descriptors into VideoLLMs, improving reasoning for video captioning, VQA, and stylistic plagiarism detection.

EDUCATION

Doctor of Philosophy in Computer Science 2021 - anticipated graduation 05/2027
University of Maryland, College Park, United States

- ◇ *Thesis: Topology-Aware Modeling of Terrain and Volumetric Data: From Discrete to Neural Implicit Representations.* GPA: 4.0/4.0

Master of Philosophy in Computer Science and Engineering 2018 - 2020
Hong Kong University of Science and Technology, Hong Kong, China

- ◇ *Thesis: Linear structure vectorization in large-scale landscape point cloud.* GPA: 3.9/4.3

BEng. in Computer Science Engineering and Electronic and Computer Engineering 2014 - 2018
Hong Kong University of Science and Technology, Hong Kong, China

- ◇ *Related coursework:* Discrete Math Tools, Advanced Computer Graphics, Data Visualization, Probability and Random Processes in Engineering. GPA: 3.9/4.3

PUBLICATIONS

- [1] **Feng, H., Aldana, D., Novello, T., & De Floriani, L. (2026). SASNet: Spatially - Adaptive Sinusoidal Networks for INRs.** CVPR 2026. [arXiv:2503.09750](https://arxiv.org/abs/2503.09750)
- [2] **Feng, H., Song, Y., & De Floriani, L. (2026). A Parallel Scale-Space Method for Critical Features Tracking on Triangulated Irregular Networks.** ACM TSAS. [doi:10.1145/3812549](https://doi.org/10.1145/3812549)

- [3] **Feng, H., Song, Y., & De Floriani, L. (2024). Critical Features Tracking on Triangulated Irregular Networks by a Scale-Space Method. SIGSPATIAL 2024. Best paper runner-up.** doi:10.1145/3678717.3691218
- [4] **Feng, H., Musunuri, S. H., & Su, G.-M. (2026). Geometry - Guided Camera Motion Understanding in VideoLLMs. CVPR 2026 Workshop PVUW.** arXiv:2603.13119
- [5] **Feng, H., Xu, X., & De Floriani, L. (2026). ImplicitTerrainV2: Wavelet - Guided Spatially Adaptive Neural Terrain Representation.** arXiv preprint. arXiv:2605.22556
- [6] **Feng, H., Xu, X., & De Floriani, L. (2026). Rethinking Amortized Neural Representations for High-Resolution Terrain Elevation Data.** arXiv preprint. arXiv:2606.00404
- [7] **Feng, H., Xu, X., & De Floriani, L. (2024). ImplicitTerrain: a Continuous Surface Model for Terrain Data Analysis. CVPR 2024 Workshop INRV.** arXiv:2406.00227
- [8] Aldana, D., Lima, J. P., Csillag, D., Perazzo, D., **Feng, H.**, Velho, L., & Novello, T. (2025). **Adaptive Training of INRs via Pruning and Densification.** arXiv preprint. arXiv:2510.23943
- [9] Zhen, M., Li, S., Zhou, L., Shang, J., **Feng, H.**, Fang, T., & Quan, L. (2020). **Learning Discriminative Feature with CRF for Unsupervised Video Object Segmentation.** ECCV 2020.

RESEARCH PROJECTS

Analytical Neural Representations of Geospatial Data

2021 - Present

University of Maryland, College Park

Advisor: Prof. Leila De Floriani

- ◇ **Topological Feature Tracking on Triangulated Irregular Networks (TINs):** Accelerated triangular mesh smoothing roughly $100\times$ with custom GPU kernels in PyTorch and optimized C++ adjacency structures, and designed an adaptive scale-space algorithm that tracks topological features on TINs at higher accuracy and lower memory cost than prior regular-grid methods. *Best paper runner-up at ACM SIGSPATIAL 2024; journal extension in ACM TSAS.*
- ◇ **Implicit Neural Representations for Terrain Surface Modeling:** Reduced terrain model storage to 25% of baseline and cut training time $4\times$ through a progressive training strategy while preserving reconstruction accuracy, and performed topological analysis and topographical feature extraction directly on the learned implicit surface function. *Published at CVPR 2024 Workshop INRV; extended in two follow-up preprints.*
- ◇ **Survey on Neural Representations of Geospatial Data:** Surveying implicit and explicit neural representations for geospatial storage, rendering, and analysis, assessing scalability, parallel computation, and support for physical simulation.

SKILLS

- ◇ **Languages:** Python, C++, C, CUDA, MATLAB, SQL, JavaScript.
- ◇ **ML & CV:** PyTorch, JAX, deep learning, implicit neural representations (INR), neural rendering, 3D Gaussian splatting, 3D reconstruction, geometry processing, point cloud processing, topological data analysis, vision-language models (VLM), multi-GPU distributed training.
- ◇ **Tools:** OpenCV, Matplotlib, MeshLab, ParaView, QGIS, W&B, UE5, Blender; Linux, SLURM, HPC.
- ◇ **Dev frameworks & scaffolds:** agentic development pipelines, MCP tool integration, custom skills/plugins, and multi-agent research orchestration.
- ◇ **Data & web:** SQL and NoSQL databases (MySQL, MongoDB), interactive data visualization (D3.js, Tableau), and full-stack web (React, Node.js).

AWARDS & HONORS

- ◇ Jacob K. Goldhaber Travel Grant and CS Department Travel Award, UMD 2026
- ◇ NSF Travel Grant, ACM SIGSPATIAL 2024
- ◇ Chair's Graduate Fellowship 2021 - 2023
- ◇ Postgraduate Studentship 2018 - 2020
- ◇ University's Scholarship Scheme for Continuing Undergraduate Students 2015 - 2018